

Hongkang Chu

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Multimodal learning; AI agents; synthetic image and video detection; MRI and MRSI reconstruction

Education

University of the Chinese Academy of Sciences 2025–present

Master’s student, Computer Science. Expected graduation: 2028.

Shanghai Ocean University 2021–2025

Bachelor’s degree, Computer Science and Technology. Undergraduate research in protein language models.

Research Experience

Chinese Academy of Sciences / UCAS 2025–present

Graduate Research

Research in MRI, MRSI, and deuterium metabolic imaging at the Innovation Academy for Precision Measurement Science and Technology.

Zhejiang University Apr–Sep 2026

Research Collaboration

Collaborative research in detecting AI-generated images and videos.

University of Illinois Urbana-Champaign Feb 2026–present

Remote Research Intern

Research in multimodal models and medical AI.

Shanghai Ocean University 2021–2025

Undergraduate Research

Protein language models for druggable protein classification.

Selected Publications and Conference Contributions

TRACE: Trajectory Representation and Consistency Estimation for AI-Generated Video Detection

ICLR 2027. Under Review. Second Author.

ClueAegis: Heuristic-to-Reasoning Cognitive-skill Learning for Unified Evidence-based Synthetic Image Detection

Findings of EMNLP 2026. Accepted. Second Author.

Physics-embedded CycleGAN for robust metabolite mapping in short-readout deuterium metabolic imaging

Magnetic Resonance Letters, 2026. Published Online. Co-author.

Phase Model-Driven Deep Learning for Robust Phase Correction in High-Throughput NMR-Based Metabolomics

The Journal of Physical Chemistry Letters, 2026. Published. Co-author.

Deep Learning Enhanced Weighted Average CSI for High-Resolution Deuterium Metabolic Imaging

ISMRM 2026. Power Pitch. First Author.

Comprehensive Research on Druggable Proteins: From PSSM to Pre-Trained Language Models

International Journal of Molecular Sciences, 2024. Published. First Author.